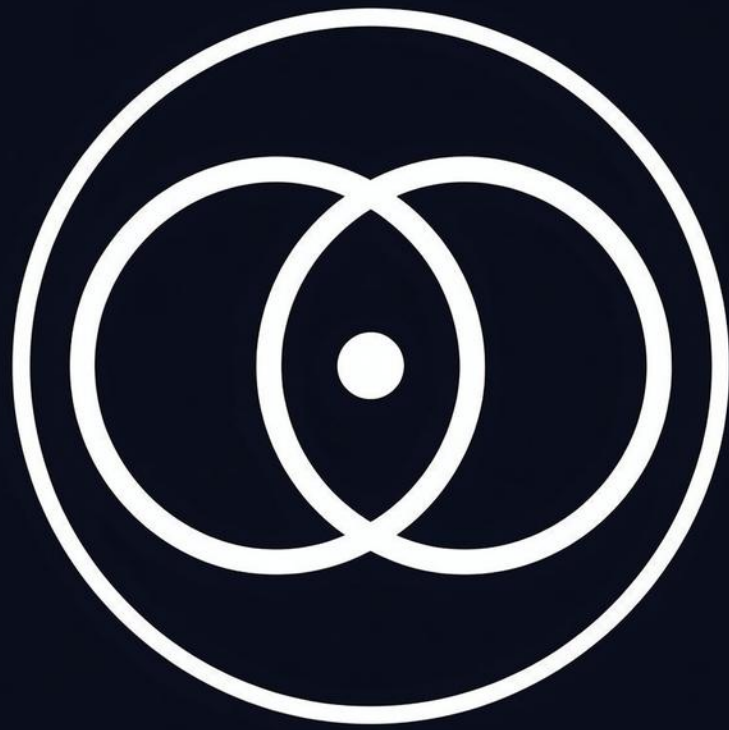




P33

The v7.6.0 Master Architecture: Unified Specifications and Backward-Compatible Rollout of Eight Structural Changes to the UDOS Engine

v7.6.0 架构总纲：八项结构变更规格

The v7.6.0 Master Architecture: Unified Specifications and Backward-Compatible Rollout of Eight Structural Changes to the UDOS Engine

Whole-volume closing master outline P33 • consolidating the eight structural upgrades across the seventeen papers P16–P32 into one unified change specification • compatible with the v7.5.0 master • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: v7.6.0 is not "yet another system swap," but **eight structured incremental changes** on the v7.5.0 engine — upgrading stop decisions from binary to three-level, adding a least-privilege scope to the handoff contract, changing the data flywheel into a breadth–depth two-stage process, adding a test-assertion gate to structural changes, and uniformly registering the four external evidences of external literature plus one "rules-first" red line into the change ledger. This paper writes these eight as landable field/state-machine/gate specifications and **repeatedly declares: they are design proposals • to be implemented, not measured returns already run.**

Evidence scope (stated only once in the whole paper): this paper is an architecture-specification document and **contains no new UDOS measured figures**. The motivation for the eight changes points item by item back to external papers (existence verified on 2026-09-21 per EVIDENCE_LEDGER_v76.md), where the performance figures of external papers are uniformly "paper-reported/company-disclosed, not independently reviewed"; existing figures on the UDOS side only point back to the v7.5.0 papers and `udos-engine/reports7/*.json` (verified / `cpu-proto` / `cpu-proxy` / unverified). All fields, state machines, gates, and thresholds this paper proposes are **design proposals • to be implemented**, and any expected return without experimental support is marked `[RESULT NEEDED]`, forbidden to be written as a measured conclusion. Verification date 2026-09-21.

Abstract

The v7.5.0 engine has answered the master proposition "structural priors determine the capability ceiling" (P0–P15), but its engineering interfaces remain **binary, single-point, ungated**: stop decisions have only the two states STOP/CONTINUE (P2), the handoff contract has only the owner field (P10), the data flywheel only asks "where has not yet been seen" (P4), and structural changes rely on people remembering to write tests (the P12 version-number incident is the counterexample). A batch of external works in the first half of 2026 (H-CSC, ClawArena-Team, RSIAgent, Ouroboros, OASIS, MAPF-GPT-DDG) exposed the roughness of these interfaces from different directions and also gave borrowable structural solutions. v7.6.0 consolidates these lessons into eight **backward-compatible structural changes**: ① three-level typed finality (P16), ② TransferBundle adds a scope field (P17), ③ BRS+DRS two-stage memory flywheel (P18), ④ structural-change review gate (P19), ⑤ external-verification registration of the heterogeneity dividend (P20, citing OASIS), ⑥ downgraded citation of MAPF million-level conclusions (P21, downgraded per ledger A8), ⑦ explicit marking of RSI lineage levels (P24, none called L5), ⑧ the design red line of rules over learning (P32). This paper gives for each: motivation source, pseudocode-level specifications of fields/state machines/gates, the compatibility path with v7.5.0, and the to-be-implemented measurement items `[RESULT NEEDED]`. The whole paper creates no new measured figures; the eight changes form a unified change set of "design proposals • to be implemented," whose success or failure is scored afterward by the five falsifiable conditions F6–F10 registered in P34.

Keywords: architecture specification; typed finality; least privilege; data flywheel; review gate; backward compatibility; design proposal; falsifiability

Structured abstract (background problem → method/thesis → evidence → contribution)

- **Background problem:** v7.5.0 proved "structure determines the ceiling," but its own interfaces remain coarse-grained, without least privilege, without change gates; the 2026 external papers both verified UDOS's core proposition and exposed the cost of making interfaces coarse.
 - **Core thesis:** v7.6.0 does not rewrite the engine, only makes **eight backward-compatible structural increments**; each increment traces to a specific finding of an external paper and has a clear old-version default backstop (default falls back to v7.5.0 semantics).
 - **Evidence clues:** the eight motivations come respectively from H-CSC v2's negative result (lexical predicates beat learned encoders), ClawArena-Team's permission bottleneck, RSIAgent's broad-then-deep strategy, Ouroboros's reviewed commits, OASIS/P9/TUMIX's heterogeneity comparison, MAPF-GPT-DDG's extrapolation boundary, and Theseus's RSI discrimination criteria; all marked per the ledger's three states.
 - **Contribution:** a unified change specification at field/state-machine/gate level; a v7.5.0→v7.6.0 change-comparison table; a discipline of "new-architecture returns uniformly marked to-be-implemented"; and the interface with P34's forecast updates (F6–F10).
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1 Why v7.6.0 is "incremental," not a "rewrite"

Reading the seventeen v7.5.0 papers (P0–P15) yields a counter-intuitive but important conclusion: the **theoretical skeleton** of the UDOS engine does not need changing. Structural priors determine the capability ceiling (P0b), hierarchical hybrid topology presses fan-in from $\mathcal{O}(N)$ to $\mathcal{O}(\log N)$ (P1), equal-weight $2f+1$ BFT-lite stop decisions (P2), homogeneous-expansion saturation (P9), and six-field handoff-contract validation (P10) — these were supported in v7.5.0 by controlled comparisons or property tests.

What really needs upgrading is **interface granularity and governance**. The 2026 external works forced this judgment from five directions:

1. **Stop decisions too coarse:** H-CSC v2 (arXiv:2606.07316) points out that LLMs produce random natural language each round, and binary STOP/CONTINUE compresses "how semantically consistent" into one switch, losing gradient information.
2. **Handoff permissions too loose:** ClawArena-Team (arXiv:2606.31174) finds the first bottleneck of multi-agent orchestration is not "cannot understand" but "over-authorization."
3. **Data collection asks only breadth:** RSIAgent (arXiv:2609.15364) uses "broad-then-deep" to prove that only covering blind spots without focusing on hard cases means memory compound interest cannot roll up.
4. **Structural changes ungated:** Ouroboros (arXiv:2608.08311) proves "reviewed commits" must become the foundation of subsequent runtime; conversely the P12 version-number incident is precisely the counterexample of "no review gate."

5. **External narratives need discipline:** the "million-level" narratives of OASIS, MAPF, Warp-Cortex must distinguish "measured" from "theoretical extrapolation/media retelling" per the evidence ledger, otherwise engineering ambition is disguised as engineering fact.

Therefore v7.6.0's positioning is: **retain all v7.5.0 behavior, overlay optional structures at eight interface points, and ensure default falls back to old behavior.** This is consistent with P32's "rules as base, learning as supplement" orientation — not overturning verified old paths, only adding guardrails on them.

Figure 1: v7.5.0 and v7.6.0 eight-change comparison

Figure 1 From v7.5.0's coarse-grained interfaces (left column) to v7.6.0's structured increments (right column), the eight changes correspond one to one. Conceptual schematic, not measured data.

2 Specifications of the eight structural changes

The eight below are uniformly given in a four-part set: **motivation (which finding of which external paper) → structural-change specification (pseudocode-level description of fields/state machines/gates) → compatibility path (how default falls back to v7.5.0) → [RESULT NEEDED] to-be-implemented items.** The full argument for each is in the corresponding dedicated paper (P16–P32); this paper only does unified registration and specification closure.

2.1 Change ①: three-level typed finality (P16)

Motivation: H-CSC v2 (arXiv:2606.07316, retitled Certifiable Semantic Agreement... What the Admissibility Instrument Decides) uses typed commit objects to solve "LLMs produce random natural language, unable to agree at byte level"; its v2 also **actively retracts** v1's "honest-retention advantage" claim (aggregate distance 5.32°/5.48° vs the steelman baseline 4.27°/4.32°, honest retention 0.6475 vs 0.7325; all paper-reported, not independently reviewed) and finds deterministic lexical predicates in adversarial scenarios at AUROC 0.865–0.982 outperform learned encoders at 0.621–0.744.

Specification (stop-decision state machine): upgrade v7.5.0 P2's binary {STOP, CONTINUE} to the three states {strong_stop, weak_stop, abort}:

- **strong_stop**: $\geq 2f+1$ members agree on the **semantic core** (judged by deterministic lexical/rule predicates, not learned encoders);
- **weak_stop**: $\geq 2f+1$ members agree only on the **verdict label** (pass/fail);
- **abort**: neither level reached, triggers view change (using P2's existing mechanism).

Pseudo state transitions:

```
round r vote -> rule predicate judges semantic-core agreement?
yes ( $\geq 2f+1$ )          -> strong_stop  commit semantic summary
no but label agree ( $\geq 2f+1$ ) -> weak_stop  commit only verdict
no                    -> abort          view change / hand to human
```

Compatibility path: old work orders and old tests default to mapping "recognize only strong_stop, weak_stop treated as CONTINUE, abort treated as failure," i.e. behavior degenerates to v7.5.0 binary semantics; new work orders explicitly declare `finality_mode=typed` to enable three states.

To implement: [RESULT NEEDED: H3-new monotonicity, weak_stop unbounded-loop risk, honest-retention difference under steelman comparison].

2.2 Change ②: TransferBundle adds a scope field (P17)

Motivation: ClawArena-Team (arXiv:2606.31174, existence verified) reports the management bottleneck of multi-agent orchestration concentrates on permission granting, with cost approximately decoupled from management quality; its precise thresholds ("no LLM exceeds 50% permission accuracy," "cost hundred-fold/total score four-fold," "SMS three-term product formula") were not hit verbatim at abstract level, and this paper only borrows the direction, not hardcoding figures.

Specification (handoff-contract new field): alongside v7.5.0 P10's six-field TransferBundle, add `scope` :

- `owner` : who is responsible (old field, retained);
- `scope` : the **set of stages this owner is allowed to operate** (new field).

Two invariants:

- On `advance` : when the current stage completes, `scope` automatically tightens to the singleton set of the next stage;
- On `handoff` : `scope` is re-granted by the new owner's authorized range; calls exceeding `scope` are directly rejected by rule predicates (`scope_violation`).

Compatibility path: default `scope=all` (i.e. no tightening), behavior equivalent to v7.5.0 with only owner; explicit `scope=stageset` enables least privilege. All 26 existing contract tests continue to pass.

To implement: [RESULT NEEDED: scope factorial experiment — boundary-violation rejection rate, usability loss, false-tightening rate].

2.3 Change ③: BRS+DRS two-stage memory flywheel (P18)

Motivation: RSI-Agent (arXiv:2609.15364) achieves training-free RSI with "broad-then-deep" memory construction, reported to make open-source Kimi-K3/GLM-5.3 surpass GPT-6-class closed-source models on specified benchmarks (paper/media reported, not independently reviewed); its "in New Environments" means learning in new environments, not cross-environment transfer.

Specification (two-stage collection strategy): upgrade v7.5.0 P4's coverage-aware single-stage flywheel to two stages sharing the same memory body:

- **BRS (breadth range scan):** use submodular greedy $\$(1-1/e)\$$ approximation to maximize state-space coverage;
- **DRS (depth range scan):** focus on identifiability blind spots — specifically pointing to hard-to-identify parameters like the "acceleration scalar skill ≈ -0.508 " already identified in P6 (single seed, cpu-proxy, not recomputed here).

Scheduling rule (design parameter): initial about 60/40 (BRS/DRS compute ratio), tilting toward DRS after coverage crosses threshold τ_{cov} .

Compatibility path: the old flywheel remains "full coverage-aware"; two stages are a new switch `flywheel=two_stage`, default off.

To implement: [RESULT NEEDED: τ_{cov} calibration, 60/40 ratio, blind-spot coverage improvement].

2.4 Change ④: structural-change review gate (P19)

Motivation: Ouroboros (arXiv:2608.08311) makes "reviewed commits" the subsequent runtime foundation, self-reporting Terminal-Bench 2.1 at 86.74% (original 86.97%, self-reported SOTA, not independently reviewed); the UDOS-side counterexample is the "version-number incident" recorded in P12 — structural changes merged without independent tests.

Specification (merge gate): any commit changing `MatrixConfig` parameters must simultaneously meet three conditions to merge:

1. commit message tagged `[structural]`;
2. accompanied by **independent test assertions** (not reusing old assertions);
3. all CI green + new assertions cover the changed structure.

Any unmet condition blocks merge.

Compatibility path: the gate only blocks structural commits tagged `[structural]`; pure documentation/format changes are unrestricted, and old repository history is not retroactively traced.

To

implement:

[RESULT NEEDED: the gate's interception rate for "silent failures/version-number-incident class," the cost to merge speed].

2.5 Change ⑤: external-verification registration of the heterogeneity dividend (P20, citing OASIS)

Motivation: OASIS (arXiv:2411.11581, NeurIPS 2024) observes with million-level **heterogeneous** social individuals "larger groups → stronger group dynamics/more diverse opinions"; this complements v7.5.0 P9's "homogeneous copies saturate no matter how many ($\text{mse}(k)=a+b/k$, cpu-proto)": P9 uses homogeneous copies to test saturation, OASIS uses heterogeneous individuals to test diversity.

Specification: this is a **citation registration and conclusion reinforcement**, not changing code interfaces. Register in the P9/P20 discussion: the quality variable is the independent-source count $\$D$, not headcount $\$N$; Warp-Cortex's Singleton Weight Sharing is in essence forced homogenization, solving only the memory wall, not the quality wall (see P22).

Compatibility path: pure documentation/argument-layer increment, no interface change, zero compatibility risk.

To implement: [RESULT NEEDED: retest the P9 saturation law inside UDOS with homogeneous vs heterogeneous copy groups].

2.6 Change ⑥: downgraded citation of MAPF million-level conclusions (P21)

Motivation: the circulated "MAPF-GPT-DDG coordinates 1M agents on a 2048×2048 map, 99.9%, $163\mu\text{s}/\text{agent}$ " was verified per ledger A8 as **only seen in Russian media retelling**; the real paper Advancing Learnable MAPF Solvers with Active Fine-Tuning (arXiv:2506.23793, IROS 2025) only verified DDG active fine-tuning and predecessor small maps, 16–32 agents measured.

Specification: v7.6.0 whole volume **uniformly downgrades the scope** — only cite "DDG active fine-tuning + large-scale coordination direction," mark "million success rate/ $163\mu\text{s}$ " as media claim or deprecated; and clarify the extrapolation boundary: MAPF is geometric, fully observable, task-structured, while UDOS work orders are semantic, contain hidden ground truth and Byzantine faults, and the two cannot directly transfer (see P21).

Compatibility path: pure citation discipline; old P1 protocol-layer conclusions ($O(N)$ to $O(\log N)$, verified extrapolation) unchanged, only adding an "end-to-end evidence not extrapolatable" disclaimer.

To implement: [RESULT NEEDED: if the MAPF paper's body later discloses million-level measurements, re-evaluate the extrapolation boundary].

2.7 Change ⑦: explicit marking of RSI lineage levels (P24)

Motivation: Theseus Labs (arXiv:2609.11873) gives seven RSI discrimination criteria and HCI; v7.5.0 P14/P15 already have the L1–L5 ladder. Externally any "gets better with use" is often called RSI, and a strict ruler is needed to place coordinates.

Specification: v7.6.0 in RSI discussion forces explicit level marking: RSI_{Agent}=L2 (changes memory/Harness), Ouroboros=L2 (review-driven Harness change), Dream-RSI=L2–L3 (optimizes exploration-strategy structure, not weights), Gödel Agent $\approx$ L4 (changes self-referential source, secondary-material scope), HyperAgents closest to "recursive meta-improvement" but **did not reach L5**, RSIAI0 (correctly recording Zenodo 15644670)=L4 candidate. **None called L5.**

Compatibility path: pure terminology discipline; does not affect engine behavior.

To implement: [RESULT NEEDED: if a system later shows controlled public evidence of "successor re-entering the loop," upgrade the level per F24-b].

2.8 Change ⑧: the design red line of rules over learning (P32)

Motivation: H-CSC v2's negative result (lexical predicates AUROC 0.865–0.982 beat learned encoders 0.621–0.744) holds in adversarial scenarios, suggesting "using learned semantic-judgment modules for safety verification" may be the wrong tool.

Specification: uniformly establish rules at the three safety-critical verification points P2/P10/P12 — **rules as base, learning as supplement:** first hold the bottom line with explainable, reproducible, CI-able deterministic rule predicates; learned models can only supplement after rules have held, and must pass steelman comparison to enter the safety path. P10's six-field mechanical validation and P2's $\$2f+1\$$ quorum judgment both remain pure rules.

Compatibility path: existing rule predicates untouched; "learning as supplement" is a new optional layer, default off.

To implement: [RESULT NEEDED: the AUROC difference between rule predicates and learned predicates at each verification point under adversarial attack (see P34 F10)].

3 v7.5.0 → v7.6.0 change-comparison table

Table 1 Quick reference for the eight structural changes (all design proposals • to implement)

#	Change point	v7.5.0 status	v7.6.0 specification	Motivation source (external paper)	Compatibility default	To-implement measurement
①	Stop decision	Binary STOP/CONTINUE	strong_stop / weak_stop / abort three states	H-CSC v2 (arXiv: 2606.07316)	Recognize only strong_stop, degenerates binary	H3-new monotonicity, unbounded loop [RESULT NEEDED]
②	Handoff contract	Only owner	owner + scope (advance tightens, handoff re-grants)	ClawArena-Team (arXiv: 2606.31174)	scope=all, equivalent old behavior	Boundary rejection rate, usability loss [RESULT NEEDED]
③	Data flywheel	Single-stage coverage-aware	BRS breadth + DRS depth two stages	RSIAgent (arXiv: 2609.15364)	flywheel default single stage	Blind-spot coverage, ratio calibration [RESULT NEEDED]
④	Structural-change governance	People remember tests	[structural] gate + independent assertions	Ouroboros (arXiv: 2608.08311)	Only blocks [structural] commits	Silent-failure interception rate [RESULT NEEDED]
⑤	Heterogeneity argument	P9 homogeneous saturation (internal)	Register OASIS heterogeneous diversity as external corroboration	OASIS (arXiv: 2411.11581)	Pure argument layer	Homogeneous vs heterogeneous group comparison [RESULT NEEDED]
⑥	Million-level citation	Directly cite MAPF million	Downgrade to DDG direction + extrapolation boundary	MAPF-GPT-DDG (arXiv: 2506.23793)	Only changes citation scope	Await paper body's million measurements
⑦	RSI terminology	L1-L5 ladder (P14/P15)	Six systems placed one by one on L1-L5, none L5	Theseus (arXiv: 2609.11873)	Pure terminology discipline	Await new "successor re-entry" evidence
⑧	Safety verification	P2/P10/P12 rules	Rules as base, learning as supplement, must pass steelman comparison	H-CSC v2 negative result	Existing rules untouched	Rules vs learning AUROC difference [RESULT NEEDED]

Table 2 Dependencies and orthogonal relations among changes

Change	Depends on	Orthogonal to	Note
① Three-level finality	Inherits P2	⑧ rules-first (uses predicates to judge semantic core)	① and ⑧ must be introduced in the same batch, otherwise strong_stop may misuse learned encoders
② scope	Inherits P10	⑧ (scope_violation itself is a rule predicate)	Independently launchable
③ Two-stage flywheel	Inherits P4/P6	①②④	Data layer, orthogonal to consensus layer
④ Review gate	Inherits P12	All	Governance layer, independently launchable first
⑤⑥⑦⑧	Pure argument/discipline	—	No runtime dependency, documentation layer

There is only one key constraint on dependencies: **① and ⑧ must be in the same batch**. If three-level finality is launched first without establishing the "rules as base" red line, engineers may well use learned encoders to judge strong_stop, precisely stepping on H-CSC v2's retraction lesson. This is the only strong coupling among the eight changes.

4 Compatibility-path overview

v7.6.0's compatibility strategy can be summarized in three sentences:

1. **Default is the old version:** all new interfaces have a default value falling back to v7.5.0 semantics (finality_mode=binary, scope=all, flywheel=single, no [structural] marking, rule predicates unchanged). Without opening new switches, behavior is byte-for-byte identical to v7.5.0.
2. **Switches made explicit:** new behavior must be opened by explicit configuration, producing no new assertions, branches, or attack surfaces before opening.
3. **Old tests do not regress:** v7.5.0's verified tests (P2's ten property tests, P10's twenty-six contract tests) fully regress in v7.6.0, ensuring increments do not break existing stock.

This interlocks with P19's review gate: any commit "changing default behavior" itself must be tagged [structural] and accompanied by assertions. That is, **v7.6.0 uses the gate it designed to constrain its own release** — a direct response to the P12 version-number incident.

5 Defects, failure boundaries, and falsifiable conditions

5.1 This master outline's own defects and boundaries

- **Not one of the eight changes has been measured effective.** This paper is a specification, not a scorecard. Reading "design proposals • to implement" as "already upgraded" is this volume's greatest misuse risk.
- **Backward compatibility ≠ zero risk.** The default backstop only guarantees "no change without opening," not "definitely better once opened"; weak_stop's unbounded loop, scope false-tightening hurting usability, and gates lowering iteration speed are all real costs (see dedicated chapters).

- **External motivation is only motivation, not evidence.** The figures of H-CSC, ClawArena, RSIAgent are mostly paper-reported/not independently reviewed; they can only show "others met problems in this direction," not substitute for UDOS's own steelman comparison.

5.2 Falsifiable conditions (aligned with P34's F6–F10)

This paper's eight specifications should be overturned or rolled back if the following occur:

- **C1 (three-level finality ineffective):** if under fault-injection comparison, the three states show no statistically measurable improvement in closure rate/false-stop rate versus binary stopping, then ① should roll back to binary.
- **C2 (scope instead worse):** if scope tightening makes normal-work-order usability loss exceed the boundary-violation rejection return, then ② should change to "on-demand authorization" rather than "default tightening."
- **C3 (two-stage flywheel no gain):** if DRS depth probing shows no improvement over BRS-only in identifiability blind-spot coverage, then ③ should fall back to single stage.
- **C4 (gate idling):** if the review gate neither blocks silent failures nor slows merge unacceptably, then ④ should be downgraded to "prompt" rather than "block."
- **C5 (rules red line pierced by counterexamples):** if on some attack family learned predicates are stable and substantially better than deterministic predicates, then ⑧ "rules as base" needs weakening to "route by attack family."

6 Conclusion

All v7.6.0 increments are in essence making **one honest remedial lesson in interface granularity and governance** on the proposition "structural priors determine the ceiling" already verified in v7.5.0: stopping must be subdivided finer, permissions tightened more, data collected deeper, changes reviewed stricter, external citations marked clearer, and safety paths held harder. Not one of the eight changes overturns v7.5.0; all are backward-compatible with default backstops; the only strong coupling is that "three-level finality" must be introduced in the same batch as "rules as base," otherwise one falls a second time in the same pit.

This is not a blueprint already redeemed, but a construction checklist with falsifiable conditions — the success or failure of each item is left to F6–F10 registered in P34 to score.

References

UDOS v7.5.0 master paper volume (this series, evidence grades marked per paper)

1. UDOS v7.5.0: P0 "Spatial Structure Itself Is Intelligence," P0b "Structure Plus Scale, Everything Obeys Scaling Laws," P0c "Metacognition and Boundary Calibration," P1 "Million-level Extrapolation of Coordination Complexity for Hierarchical Hybrid Topology," P2 "BFT-lite for Multi-Agent Stop Decisions," P4 "Coverage-aware Data Flywheel," P6 "Empirical Observability of Hidden Dynamic Parameters," P9 "The Quality Saturation Law of Agent Legions and the Failure Boundary of Homogeneous Expansion," P10 "Making TransferBundle Handoff Information Loss Measurable," P11 "Settlement Conservation in the Internal Task Market," P12 "Evidence-grade-driven Reproducible Systems Engineering," P14 "The Convergence of RSI," P15 "Life Originates in Structural Organization."

UDOS v7.6.0 dedicated papers (full argument for the eight changes)

1. P16 "Typed Finality: Three-level Semantic Commits and BFT-lite Stop-Decision Upgrade"; P17 "Least-privilege Orchestration: TransferBundle's scope Tightening and Re-granting"; P18 "Breadth-Depth Two-stage Memory Flywheel: BRS and DRS Submodular Collection"; P19 "Structural-change Review Gate: Test-assertion Enforcement for Versioned Repositories"; P20 "Heterogeneity Dividend and Homogeneity Saturation"; P21 "Million-level Coordination End-to-end Evidence and Extrapolation Boundary"; P22 "The Memory Wall Is Not the Coordination Wall"; P24 "Measured Positioning of the RSI Lineage"; P32 "Rules Over Learning"; P34 "v7.6.0 Literature Evidence Ledger and Forecast Updates."

External dialogue counterparts (existence verified on 2026-09-21 per EVIDENCE_LEDGER_v76.md; figures paper-reported/company-disclosed, not independently reviewed)

1. H-CSC Team (2026). Certifiable Semantic Agreement Among LLM Agents: What the Admissibility Instrument Decides (v2). arXiv:2606.07316.
2. ClawArena-Team (2026). Benchmarking Subagent Orchestration and Dynamic Workflows in Language-Model Agents. arXiv:2606.31174.
3. Zhu, S., et al. (2026). RSI-Agent: Autonomous Exploration for Recursive Self-improvement in New Environments. arXiv:2609.15364.
4. Ouroboros (2026). A Self-Developing Frontier Coding Agent with Reviewed Core Evolution. arXiv:2608.08311.
5. Yang, K., et al. (2024). OASIS: Open Agent Social Interaction Simulations with One Million Agents. arXiv:2411.11581, NeurIPS 2024.
6. MAPF Team (2025). Advancing Learnable Multi-Agent Pathfinding Solvers with Active Fine-Tuning. arXiv:2506.23793, IROS 2025. ("Million success rate" downgraded per ledger A8 to media claim.)
7. Theseus Labs (2026). Toward Genuine Recursive Self-Improvement. arXiv:2609.11873.

Evidence discipline and reproduction notes

- This paper is an **architecture-specification document**, with no new experiments; all eight changes are **design proposals • to implement**, all return positions [RESULT NEEDED], forbidden to be written as measured.
- The existence, titles, and arXiv numbers of external papers were verified on 2026-09-21 per EVIDENCE_LEDGER_v76.md; their performance figures (H-CSC aggregate distance/AUROC, Ouroboros Terminal-Bench, RSI-Agent open-source surpassing closed-source, MAPF million success rate, etc.) uniformly keep the scope "paper-reported/company-disclosed, not independently reviewed," and MAPF million figures are downgraded per A8 to media claims.
- UDOS existing figures only point back to v7.5.0 papers and reports7/*.json, with evidence grades per paper (verified / cpu-proto / cpu-proxy / unverified); single-seed pilots need ≥ 30 seeds and 95% CI before submission.
- This paper's falsifiable conditions C1–C5 correspond one to one with P34's F6–F10, all testable afterward with controlled comparisons within the 2027 scoring window.